Homework 1

ORF 245 - Fundamentals of Statistics

Due: Feb. 7th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Mean Absolute Deviation) Consider the following data set:

$$x_1 = 10.03$$
$$x_2 = 41.61$$
$$x_3 = 5.32$$
$$x_4 = 7.78$$
$$x_5 = 8.5$$
$$x_6 = 0.2$$
$$x_7 = 1.1$$

(a) Compute the range of $x_1, ..., x_n$.

- **Solution**:
  $\text{Range}(x_1, \ldots, x_7) = \max_i x_i - \min_i x_i = 41.41$.

(b) Calculate the sample variance and standard deviation of $x_1, ..., x_n$.

(**Note:** In this course, we've defined the sample variance as dividing by n, and not $n-1$.)

- **Solution**:
  $\sigma^2 = \sum_i (x_i - \bar{x})^2/7 = 171.5113$,
  $\sigma = \sqrt{\sigma^2} = 13.09623$.

(c) Another measure of variability, which is more robust than the standard deviation or variance, is called the *median absolute deviation*, which we'll denote by $\text{MAD}(x_1, ..., x_n)$. For a data set $x_1, \ldots, x_n$, the sample MAD is defined as:

$$\text{MAD}(x_1, ..., x_n) = \text{median}\{\ |x_i - x_{\text{med}}|, \quad i = 1, \ldots, n\}.$$

That is, MAD is the median of each data point's absolute deviation from the sample median.

Compute MAD of $x_1, ..., x_n$ above.

- **Solution**:
  $x_{\text{med}} = 7.78$,
  $\text{MAD} = 2.46$.

(d) Let $z_1, ..., z_n$ be the following slightly different data set:

$$\begin{aligned} z_1 &= \mathbf{100.3} \\ z_2 &= 41.61 \\ z_3 &= 5.32 \\ z_4 &= 7.78 \\ z_5 &= 8.5 \\ z_6 &= 0.2 \\ z_7 &= 1.1 \end{aligned}$$

Recompute the sample standard deviation and MAD of $z_1, ..., z_n$. Compare these values with your answers from parts (b) and (c), what changes do you see?

- **Solution**:
  $\sigma = 33.96107$,
  $\text{MAD} = 6.68$.

  Standard deviation is more sensitive to the changes of individual samples while MAD is more resistant to the influence of a single sample, exhibiting greater stability. Hence, indeed the MAD does seem to be more robust than the sample standard deviation.

Exercise: 2. (Additive Properties) Suppose we have two data sets $x = \{x_1, \ldots, x_n\}$ and $y = \{y_1, \ldots, y_n\}$ and let $z = x + y = \{x_1 + y_1, \ldots, x_n + y_n\}$. Recall the definitions of sample mean and variance:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad\qquad \sigma_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

(a) Prove or disprove that the mean satisfies the additive property, i.e., prove or disprove that $\bar{z} = \bar{x} + \bar{y}$.

(**Note:** to disprove the statement, it is only necessary to give a counterexample. I.e. come up with two specific datasets x and y, of any numbers and any size, for which this equality is not satisfied. The same principle applies to other "prove or disprove" problems.)

- **Solution**:

  Proof.

$$\bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i = \frac{1}{n}\sum_{i=1}^{n}(x_i + y_i) = \frac{1}{n}\sum_{i=1}^{n} x_i + \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{x} + \bar{y}.$$

$\square$

(b) Prove or disprove that the median satisfies the additive property, i.e., prove or disprove that $z_{\text{med}} = x_{\text{med}} + y_{\text{med}}$.

- **Solution**:
  Counterexample: Let $x = \{-2, -1, 0\}, y = \{2, 0, 2\}$. Then $z = x+y = \{0, -1, 2\}$, $x_{\text{med}} = -1$ and $y_{\text{med}} = 2$. We have $z_{\text{med}} = 0$, but $x_{\text{med}} + y_{\text{med}} = 1$.

(c) Give an example of two specific data sets $x_1, ..., x_n$ and $y_1, ..., y_n$ (of any numbers and any size) showing that variance does **not** satisfy the additive property. That is, disprove $\sigma_z^2 = \sigma_x^2 + \sigma_y^2$.

- **Solution**:
  Let $x = y = \{0, 2\}$. Then $\sigma_x^2 = \sigma_y^2 = 1$, $z = \{0, 4\}$, and $\sigma_z^2 = 4 \neq \sigma_x^2 + \sigma_y^2$.

Exercise: 3. (Equality of Sets) Let A, B, and C be three sets, all of which are subsets of some S.
(**Note:** You can absolutely disprove a statement using visual or numerical counterexamples, but specific examples (whether drawings or numerical) are not enough to prove a statement.)

(a) Prove: $A \cup B = (A \cap B) \cup (A \cap B^C) \cup (A^C \cap B)$

- **Solution**:
  (Part 1): Note that
  $$A \cap B \subset A \cup B, \;\; A \cap B^C \subset A \cup B, \;\; A^C \cap B \subset A \cup B.$$
  Thus, we have
  $$(A \cap B) \cup (A \cap B^C) \cup (A^C \cap B) \subset A \cup B.$$
  (Part 2): For any $x \in A \cup B$, it means it is either in A, in B, or in both.
  If x is in both A and B, it's in $(A \cap B)$.
  If x is only in A (and not in B), it's in $(A \cap B^C)$.
  If x is only in B (and not in A), it's in $(A^C \cap B)$.
  Thus we have:
  $$A \cup B \subset (A \cap B) \cup (A \cap B^C) \cup (A^C \cap B).$$

(b) Prove or disprove: $A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$

- **Solution**:
  Disprove: $A = \{1, 2\}$, $B = \{3\}$, $C = \{3\}$. Then
  $$A \cup (B \cap C) = \{1, 2, 3\}, \quad (A \cap B) \cup (A \cap C) = \emptyset.$$

(c) Prove or disprove: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

- **Solution**:
  We prove the statement in the following.
  An element x belongs to $A \cup (B \cap C)$ if: 1. x belongs to A, **or** 2. x belongs to both B and C.
  An element x belongs to $(A \cup B) \cap (A \cup C)$ if: 1. x belongs to A or B, **and** 2. x belongs to A or C.
  For any $x \in A \cup (B \cap C)$: if $x \in A$, then $x \in (A \cup B) \cap (A \cup C)$; if $x \in B \cap C$, then $x \in (A \cup B) \cap (A \cup C)$. Thus, we have
  $$A \cup (B \cap C) \subset (A \cup B) \cap (A \cup C).$$
  Similarly, we can show
  $$(A \cup B) \cap (A \cup C) \subset A \cup (B \cap C).$$

Exercise: 4. (Working with Probability Axioms) Suppose we have an unfair die where the sample space is given by $S = \{1, 2, 3, 4, 5, 6\}$, and the probabilities of each outcome are:

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \frac{1}{4},$$
$$\mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\}) = \frac{1}{12}.$$

(a) Let $A = \{1, 2, 4\}$. Compute $\mathbb{P}(A)$ and $\mathbb{P}(A^C)$.

- **Solution**:
  $\mathbb{P}(A) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(4) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} = \frac{7}{12}$, $\mathbb{P}(A^C) = 1 - \mathbb{P}(A) = \frac{5}{12}$.

(b) Let $B = \{\text{the roll is an odd number}\}$. Compute $\mathbb{P}(B)$.

- **Solution**:
  $\mathbb{P}(B) = \mathbb{P}(1) + \mathbb{P}(3) + \mathbb{P}(5) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} = \frac{7}{12}$.

(c) Let $C = \{\text{the roll is a prime number}\}$. Compute $\mathbb{P}(C)$.

- **Solution**:
  $\mathbb{P}(C) = \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(5) = \frac{1}{4} + \frac{1}{4} + \frac{1}{12} = \frac{7}{12}$.

(d) Compute $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$, and $\mathbb{P}(A \cap C)$.

- **Solution**:
  $\mathbb{P}(A \cap B) = \mathbb{P}(1) = \frac{1}{4}$,
  $\mathbb{P}(A \cup B) = \mathbb{P}(1) + \mathbb{P}(2) + \mathbb{P}(3) + \mathbb{P}(4) + \mathbb{P}(5) = 1 - \frac{1}{12} = \frac{11}{12}$,
  $\mathbb{P}(A \cap C) = \mathbb{P}(2) = \frac{1}{4}$.

Exercise: 5. [OPTIONAL] (Properties of Probability)
Note: This is an ***optional*** exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let S be the sample space of some experiment, and $\mathbb{P}$ a probability measure. Let $A \subset S$, and $B \subset S$ be any two events. Use the Axioms of Probability to prove the following properties:

(a) $\mathbb{P}(A^C) = 1 - \mathbb{P}(A)$.

- **Solution**:
  Since $S = A \cup A^c$ and $A \cap A^c = \emptyset$, we have

  $$\mathbb{P}(S) = \mathbb{P}(A) + \mathbb{P}(A^c).$$

  Since $\mathbb{P}(S) = 1$, we have $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

(b) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

- **Solution**:
  First, observe that we can write $A \cup B$ as $(A \backslash (A \cap B)) \cup (B \backslash (A \cap B)) \cup (A \cap B)$, and these three sets are disjoint from each other. Then by the axiom of probability, we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A \backslash (A \cap B)) + \mathbb{P}(B \backslash (A \cap B)) + \mathbb{P}(A \cap B).$$

  Next, observe that A can be rewritten as $A = (A \backslash (A \cap B)) \cup (A \cap B)$ and these two sets are disjoint. Again, by the axiom of probability, we have

$$\mathbb{P}(A) = \mathbb{P}(A \backslash (A \cap B)) + \mathbb{P}(A \cap B).$$

  Therefore, we have

$$\mathbb{P}(A \backslash (A \cap B)) = \mathbb{P}(A) - \mathbb{P}(A \cap B).$$

  By the similar argument, we also have

$$\mathbb{P}(B \backslash (A \cap B)) = \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

  Plugging these terms back to the first equation, we have

$$\begin{aligned} \mathbb{P}(A \cup B) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) + \mathbb{P}(B) - \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B). \end{aligned}$$

(c) $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.

(Hint: You can assume that part (b) has already been proved.)

- **Solution**:
  By non-negativity of probability, we have $\mathbb{P}(A \cap B) \geq 0$. Then by part (b), we have

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

(d) **(Challenging)** If $B_1, ..., B_n$ are mutually exclusive and collectively exhaustive, then $\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i)$.

(Hint: First, try to show that $A = \cup_{i=1}^{n}(A \cap B_i)$. Then, show that the sets $A \cap B_1$, ..., $A \cap B_n$ are mutually exclusive, and conclude using Axiom 3 of Probability.)

- **Solution**: Since $B_1, \ldots, B_n$ are collectively exhaustive, we have

$$S = \bigcup_{i=1}^{n} B_i.$$

Intersecting both sides with A gives

$$A = A \cap S = A \cap \left(\bigcup_{i=1}^{n} B_i \right) = \bigcup_{i=1}^{n} (A \cap B_i).$$

Next, because the B_i's are mutually exclusive (i.e., $B_i \cap B_j = \varnothing$ for $i \neq j$), it follows that

$$(A \cap B_i) \cap (A \cap B_j) = A \cap B_i \cap A \cap B_j = A \cap (B_i \cap B_j) = \varnothing \quad \text{for } i \neq j.$$

Thus, the sets $\{A \cap B_i\}_{i=1}^{n}$ are pairwise disjoint.
By Axiom 3,

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} (A \cap B_i)\Big) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i).$$

Since $A = \bigcup_{i=1}^{n} (A \cap B_i)$, we immediately get

$$\mathbb{P}(A) = \sum_{i=1}^{n} \mathbb{P}(A \cap B_i).$$

This completes the proof.